

GANASHREE C R

Phone: 9353293463 | Email: shreegana030@gmail.com | LinkedIn: [linkedin.com/in/ganashree-c-r-8988062a7](https://www.linkedin.com/in/ganashree-c-r-8988062a7)
Address: Chandrashekharpura, Gubbi Taluk, Tumkur District - 572213

SUMMARY:

Final-year Electrical and Electronics Engineering student with practical experience in industrial IoT monitoring and model predictive control for EV charging systems and power systems simulation using Simulink/Simscape Electrical. Skilled in embedded C/C++, Python, and communication protocols including CAN Bus, Modbus, RS-485, and LoRaWAN. Seeking a role where hardware and software intersect to build reliable, real-world systems.

EDUCATION:

Dayananda Sagar Academy of Technology and Management Bachelor of Engineering — Electrical and Electronics Engineering; CGPA: 7.5	<i>Bengaluru, India.</i> 2022–2026
Arvind PU College Pre-University — PCMB; Percentage: 65%	<i>Tumkuru, India.</i> 2020–2022
Varin International Residential School Indian Certificate of Secondary Education (ICSE); Percentage: 80%	<i>Tumkuru, India.</i> 2014–2020

TECHNICAL SKILLS:

- **Programming Languages:** C/C++, Python, Verilog HDL
- **Tools:** MATLAB, AutoCAD, Arduino IDE, Altium Designer
- **Coursework:** Analog Electronic Circuits, Digital Electronic Circuits, Power Electronics, Switchgear and Protection, Electric Motor and Drive System for Electric Vehicles, Power system analysis, Control Systems
- **Communication Protocols:** RS-485, Modbus RTU, MQTT, LoRaWAN, I²C, SPI, UART, CAN, CAN FD, LIN

PROJECTS:

- 1. GSM-Based Prepaid Energy Meter using Arduino** *October – December, 2024*
 - Built a smart prepaid energy meter that measures real-time power consumption and sends balance alerts via GSM SMS, enabling remote recharge and implemented relay-based load disconnect on zero credit without need of physical access.
 - Integrated energy consumption measurement and prepaid billing functionality with GSM communication to enhance user convenience and operational efficiency through automation and real-time monitoring.
- 2. PV-EV Charging Station via MPC** *August – December, 2025*
 - Designed a grid-connected PV-EV charging architecture where an MPC controller optimally schedules charging loads to maximise renewable utilisation and minimise grid energy cost.
 - Achieved 85%+ solar energy utilisation and 20–30% reduction in electricity cost through predictive horizon planning over a 24-hour cycle.
 - Modelled DC-DC boost converter, bidirectional inverter, and battery SoC dynamics in Simulink with closed-loop MPC reference tracking and constraint handling for grid power limits — achieving stable fast-response control with settling time under 50 ms.

EXPERIENCE:

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| Gravity India Technologies Pvt Ltd – R&D intern | <i>December 2025 – March 2026</i> |
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- Developed Building management system which is a wireless IoT monitoring system for Diesel Generators and Solar Inverters in real time using ESP32 microcontrollers, RS-485, and Modbus RTU for sensor data acquisition.
 - Implemented LoRaWAN transmission for long-range, low-power data relay from field nodes to gateway, eliminating wired infrastructure.
 - Integrated MQTT broker for cloud publish-subscribe messaging and built live dashboards with alert thresholds for voltage, current, frequency, and fault states.

CERTIFICATIONS:

- **Introduction to Motor Control — MathWorks:** Strengthened understanding of motor control systems in embedded systems, industrial automation, and electric vehicle applications.
- **Power Systems Simulation Onramp — MathWorks:** Used Simulink & Simscape Electrical to build and simulate a three-phase power system – including transformer, transmission line, and fault events – while measuring voltage, current, and power

SOFT SKILLS:

- Communication, Negotiation, Leadership and Management, Presentation Skills, Adaptability, Emotional Intelligence